Critical Blind Spots & Amendments

The following three issues were identified during roadmap review. Each requires a concrete fix before Phase 1 begins. They are addressed as direct amendments to the relevant phase task lists.

Blind Spot 1 — VMI Smart-Card / Mobile-ID Authentication

VMI (imas.vmi.lt) and Sodra (draudejai.sodra.lt) require Smart-ID or Mobile-ID for login. An autonomous agent cannot approve a push notification on your phone. Without handling this explicitly, the agent stalls silently after triggering the login step and may lock the government account after repeated failed attempts — triggering a manual investigation.

Amendment — vmi_submit.md and sodra_draft.md (Phase 3)

Add an explicit "auth pause" block to both skill files:

1. Agent navigates to the portal login page and enters the company code.
2. Agent calls notify_owner(): "Smart-ID push sent for VMI login — please approve on your phone within 60 seconds."
3. Agent polls for an authenticated session cookie for up to 90 seconds.
4. If no session confirmed within 90s: log portal_action outcome=failure, alert owner: "Login timed out — please log in manually and re-trigger the task."
5. Circuit breaker rule applies: three consecutive auth failures pause all portal tasks.

Blind Spot 2 — OCR Confidence Gate

Receipts from small Lithuanian suppliers are frequently crumpled, thermally faded, or partially handwritten. A single misread digit — "9" parsed as "4" — in a VAT amount produces an incorrect i.SAF entry that will not match the supplier's own filing, triggering a VMI cross-check flag and a potential manual audit.

Amendment — scan_receipt.py (Phase 3, Task 1)

Implement a three-tier confidence gate after the docTR / EasyOCR extraction step:

Tier 1 — Confidence ≥ 90%: proceed automatically to isaf_generator.py.

Tier 2 — Confidence 70–89%: send the receipt image to the owner via Telegram with all extracted fields pre-filled. Do not generate XML until the owner confirms or corrects.

Tier 3 — Confidence < 70%: move file to workspace/manual_review/. Alert owner with filename. Never attempt XML generation on sub-threshold data.

Store the raw confidence score in receipt_processing.ocr_raw_json for audit purposes.

Blind Spot 3 — POS Pull Model (Polling over Webhooks)

Not all Lithuanian POS systems (Paysera, RoboLabs) support real-time webhooks for every transaction event. Building the integration layer around a push model and discovering this limitation mid-Phase 2 would require a significant rewrite of the entire POS connector.

Amendment — core/integrations/ architecture (Phase 2, Task 1)

Build pull-first from day one:

1. Implement a polling loop (every 15 minutes via APScheduler) as the primary data source.
2. In Phase 4, add an optional FastAPI webhook endpoint. If the POS supports push events,
   it writes into the same internal queue the poller uses.
3. All downstream modules read only from the queue — they never care whether data arrived
   via push or pull. This makes the webhook an enhancement, not a dependency.

Revised Phase 1 task order:

Day 1 — POS API smoke test (if this fails, the audit pipeline has no data source).

Day 2 — Telegram bot setup (notification infrastructure for all subsequent debugging).

Budget Estimate

All figures are estimates in EUR. Development cost assumes a solo developer in Lithuania at a blended rate of €35/hr. A two-person team (developer + part-time tester) can compress the timeline and reduce total hours billed.

One-Time Development Cost

Phase / Item	Hrs (solo)	Hrs (2-person)	Rate €/hr	Cost €
Phase 1 — Environment & Foundation	16	8	35	560
Phase 2 — Python Deterministic Core	60	30	35	2,100
Phase 3 — OpenClaw Agent Brain	80	40	35	2,800
Phase 4 — Dashboard & Hardening	60	30	35	2,100
Phase 5 — Production Launch	24	12	35	840
Compliance review (accountant / lawyer)	—	—	—	300–500
Contingency (15%)	—	—	—	≈1,260
TOTAL (solo developer)	**240**	**120**	**35**	**≈9,660**

Monthly Operating Cost (Steady State)

Service / Item	Est. monthly (€)	Notes
OpenClaw agent usage	20–60	Depends on portal session frequency
VPS / hosting (1 vCPU, 2 GB RAM)	5–12	Hetzner / DigitalOcean / on-site Raspberry Pi
Telegram Bot API	0	Free tier is sufficient
POS API (Paysera / RoboLabs)	0–15	Check your existing POS contract
docTR OCR (self-hosted)	0	Runs on the same VPS; no external API fee
Developer maintenance (2 hrs/mo)	70	At €35/hr: schema updates, monitoring
TOTAL estimated monthly	**≈100–160**	**Excludes one-off developer spikes**

ROI Threshold

At €100–160/month operating cost, the system breaks even if it saves approximately 3–4 hours of manual bookkeeping and compliance work per month. This is a conservative estimate for a shop filing VMI monthly and managing 3–6 baristas on Sodra. Additional value accrues from avoided VMI fines (up to €4,300 per incident) and reduced accountant hours at month-end.

RACI Matrix

Defines who is Responsible (does the work), Accountable (owns the outcome), Consulted, or Informed for every key activity across the full project lifecycle and steady-state operations.

R	Responsible	A	Accountable	C	Consulted	I	Informed

Activity	Developer	Owner	Accountant	Agent
Phase 1 — Environment setup	R/A	I	—	—
Phase 2 — Python core development	R/A	I	C	—
Phase 3 — OpenClaw agent development	R/A	C	—	—
Phase 4 — Dashboard & compliance review	R	A/C	C	—
Phase 5 — Production launch	R	A	I	—
Daily Z-report audit (07:00)	I	I	I	R/A
Telegram alert: AUDIT_MISMATCH	I	A	I	R
Receipt OCR processing	I	I	I	R/A
i.SAF XML generation & XSD validation	I	I	C	R/A
VMI portal — draft filing	I	A	I	R
VMI e-Signature submission	—	R/A	I	I
Smart-ID / Mobile-ID approval	—	R/A	—	I
Sodra 1-SD form draft	I	A	—	R
Sodra 1-SD e-Signature submission	—	R/A	—	I
Inventory reorder draft	I	A	—	R
Supplier email — send (after GO)	—	R/A	—	I
Monthly P&L draft generation	I	A	R/C	R
Monthly i.SAF submission (by 15th)	—	R/A	C	I
New hire detection	I	I	—	R/A
Weekly bank reconciliation	I	A	R/C	R
Circuit breaker: portal lock alert	I	A	—	R
OCR confidence < 70%: manual review flag	I	R/A	—	I
Monthly skill file / XSD update review	R/A	I	—	—

How to read this table

R/A (teal) — agent runs this task fully autonomously.
A (navy) — owner must take an explicit action (e-sign, confirm, approve).
R (light) — responsible party executes; someone else is accountable.
— — this role has no involvement in this activity.

Critical rule: no row in the VMI or Sodra submission section has R on the Agent column. The agent drafts, alerts, and waits. The owner always holds the pen.